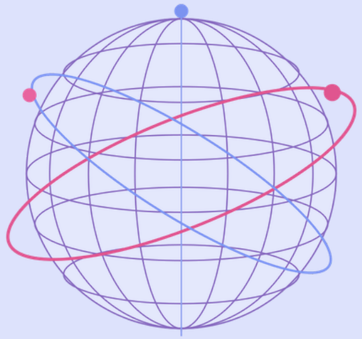
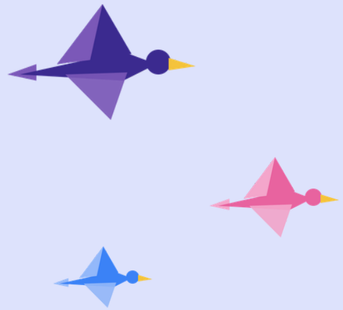
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
EntangleOps — A Research Workbench for Quantum Networks

TRACK
Quantum Communication and Networking

TEAM NAME
Quant Rangers

TEAM LEADER
Rebecca Peter Dsouza

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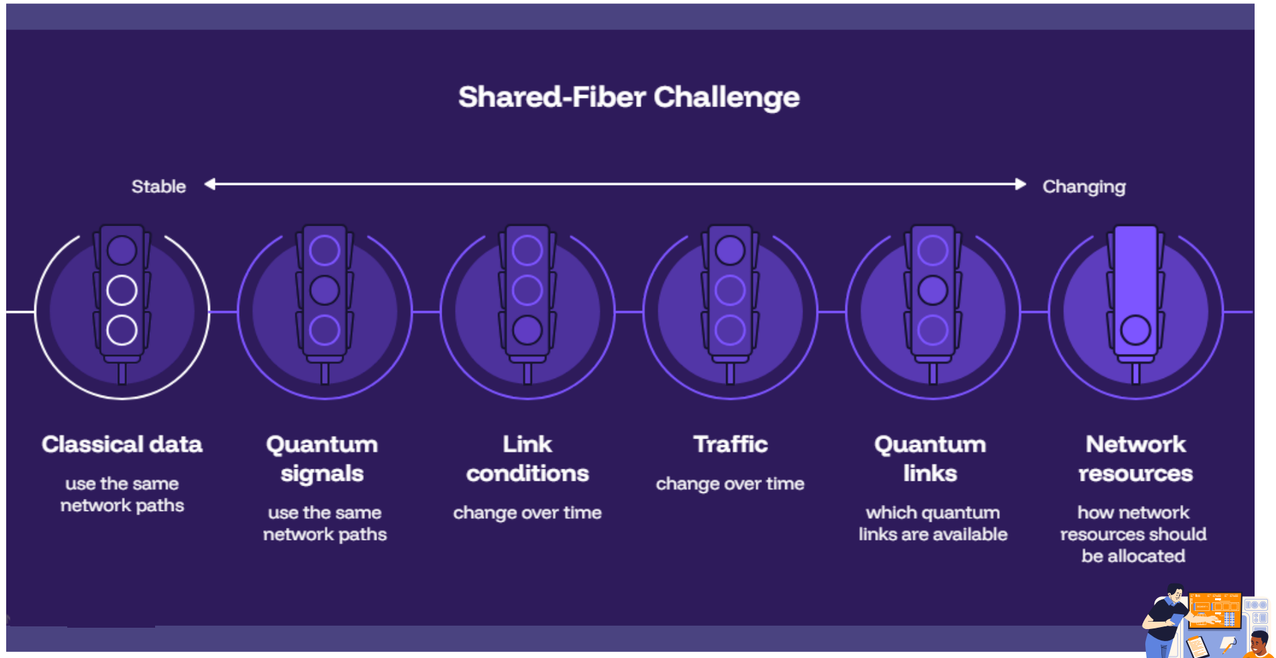
TEAM MEMBERS
Member 1: Mangirish Shyam Prabhavalkar

INSTITUTION / ORGANISATION
Kolhapur Institute of Technology

COUNTRY
India

Problem Statement

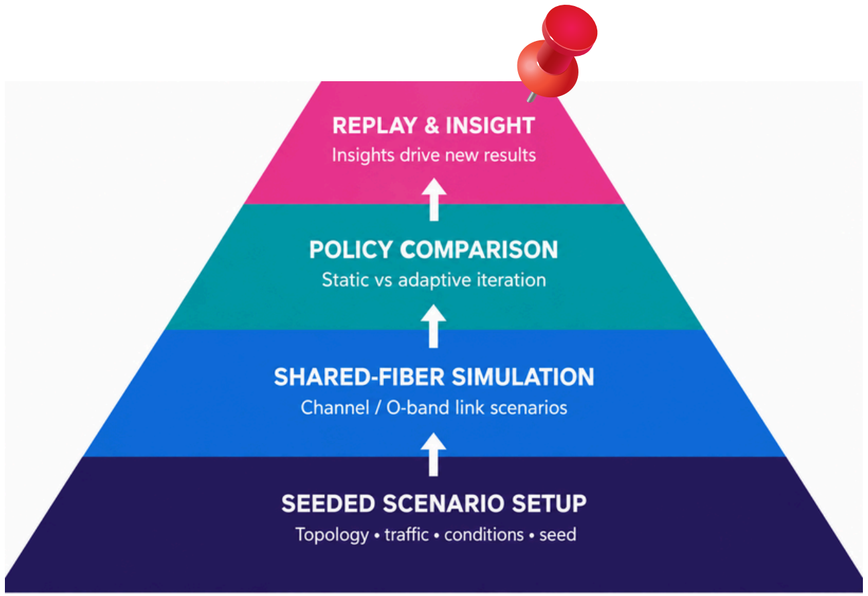
- Quantum network links share fiber with classical traffic, which can affect link quality and availability.
- Researchers need to understand how changing network conditions and competing allocation policies affect performance.
- Without a reproducible simulation workbench, it is difficult to compare scenarios consistently, track results, and inspect system behavior.
- EntangleOps addresses this by providing a software environment for repeatable quantum network simulations and policy comparisons.





Objectives & Novelty

Objectives	vs	Novelty
Build a reproducible workbench to simulate quantum networks sharing fiber with classical traffic.		Brings simulation, policy comparison, prediction, and run replay into one browser-based research workflow.
Compare static and adaptive resource allocation policies under changing traffic and link conditions. Model link behavior across C-band and O-band scenarios using seeded, repeatable experiments.		Uses repeatable scenarios to support fair comparisons across policies. Combines network-analysis features with experiment and model metadata, plus monitoring and security controls.
Record runs and results so scenarios can be inspected and replayed. Provide prediction tools for Werner-state fidelity and PTP one-way delay.		Keeps the scope practical and transparent: it is a software research simulator, not a controller for physical quantum hardware.



Solution & Quantum Advantage

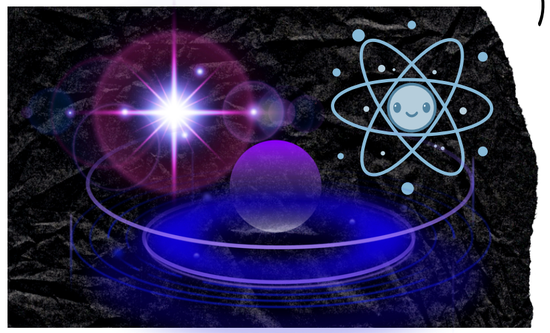
EntangleOps is a browser-based research workbench for exploring quantum networks that share fiber with classical traffic.

- Create repeatable network scenarios with seeded traffic and environmental conditions.
- Simulate link behavior across C-band and O-band.
- Compare static and adaptive allocation policies, then inspect and replay saved runs.
- Explore Werner-fidelity and PTP-delay predictions alongside run results.

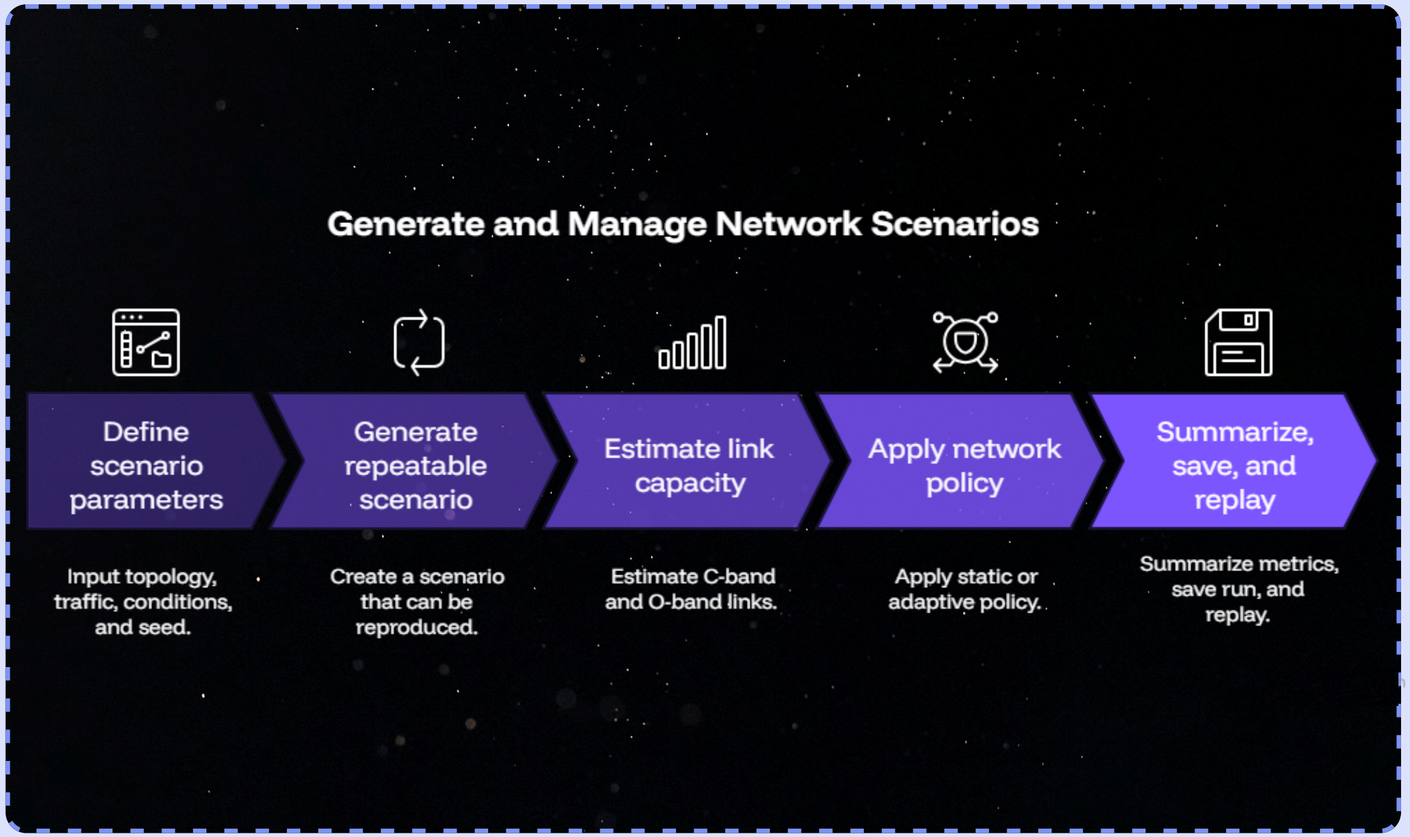
Quantum Advantage

The project does not claim a speedup from quantum computing. Its advantage is a practical research workflow for studying quantum-network behavior: reproducible scenarios make it easier to compare policies and understand how changing conditions affect link availability and performance.





Algorithm & Circuit Design



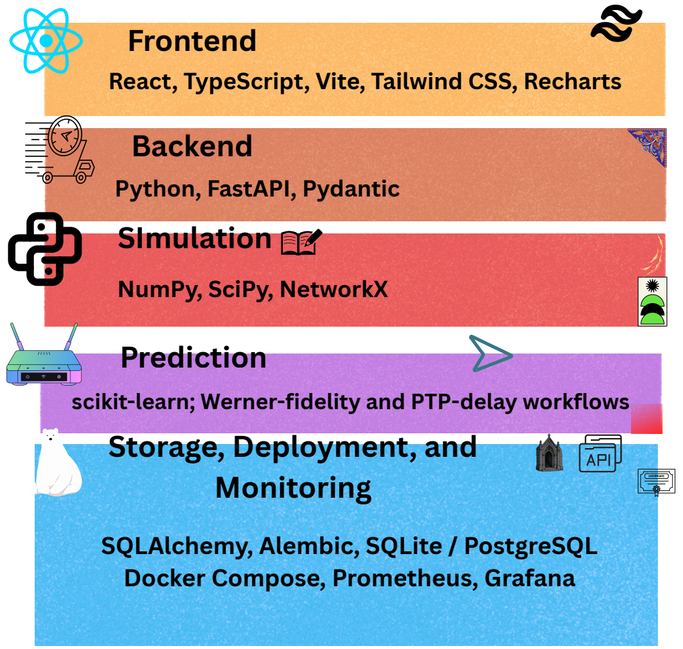
- Inputs: Metro topology, traffic profile, environmental conditions, and random seed
- Link model: Analytic estimates for C-band and O-band
- Policy step: Run static or adaptive allocation on the scenario
- Outputs: Performance summaries, time-series results, and saved run history
- Additional estimates: Werner-state fidelity and PTP one-way delay
- Implementation: FastAPI simulation service with a React browser console

Implementation & Tech Stack

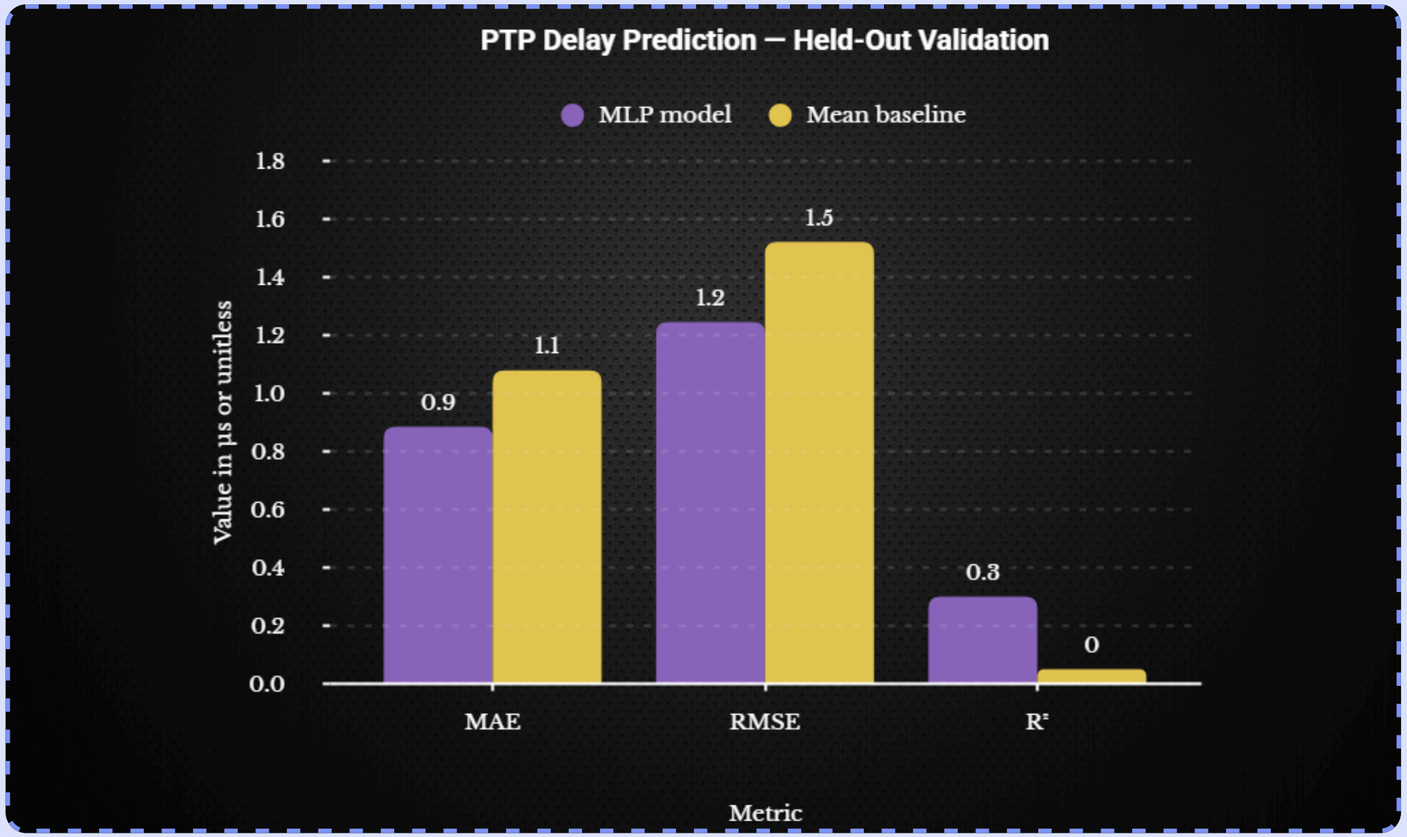
React + TypeScript Console
Users configure network scenarios, launch simulations, and explore topology, results, time-series data, and saved runs.

FastAPI Simulation Service
Validates requests, generates seeded traffic and environmental traces, estimates link behavior, runs allocation policies, and stores results for later review.

Data & Operations
SQLAlchemy with SQLite or PostgreSQL stores runs and experiment records. API-key roles, audit logging, health checks, and metrics support controlled operation and monitoring.



Results & Analysis



Made with Napkin

- Validation used 48,000 measurements from 12 held-out CSV runs.
- The model predicts one-way PTP path delay from run conditions and recent measurements.
- Lower MAE and RMSE show better prediction error than the baseline on these held-out runs.
- Results are limited to the available SLATE-PTP configurations; broader generalization has not been established.

Impact & Real-World Applications

EntangleOps provides a repeatable way to explore how classical traffic and changing link conditions may affect quantum-network links. Researchers can compare allocation policies in the same simulated scenario and review saved results, helping them identify questions to investigate before physical testing.

Network planning	Policy evaluation	Research and teaching	Telemetry analysis	Operational review
Explore link and traffic scenarios for hybrid classical; quantum fiber networks.	Compare resource-allocation approaches under changing conditions.	Demonstrate quantum-network trade-offs through reproducible simulations.	Estimate PTP one-way delay from measured run conditions and recent observations.	Retain run history and model metadata for inspection.

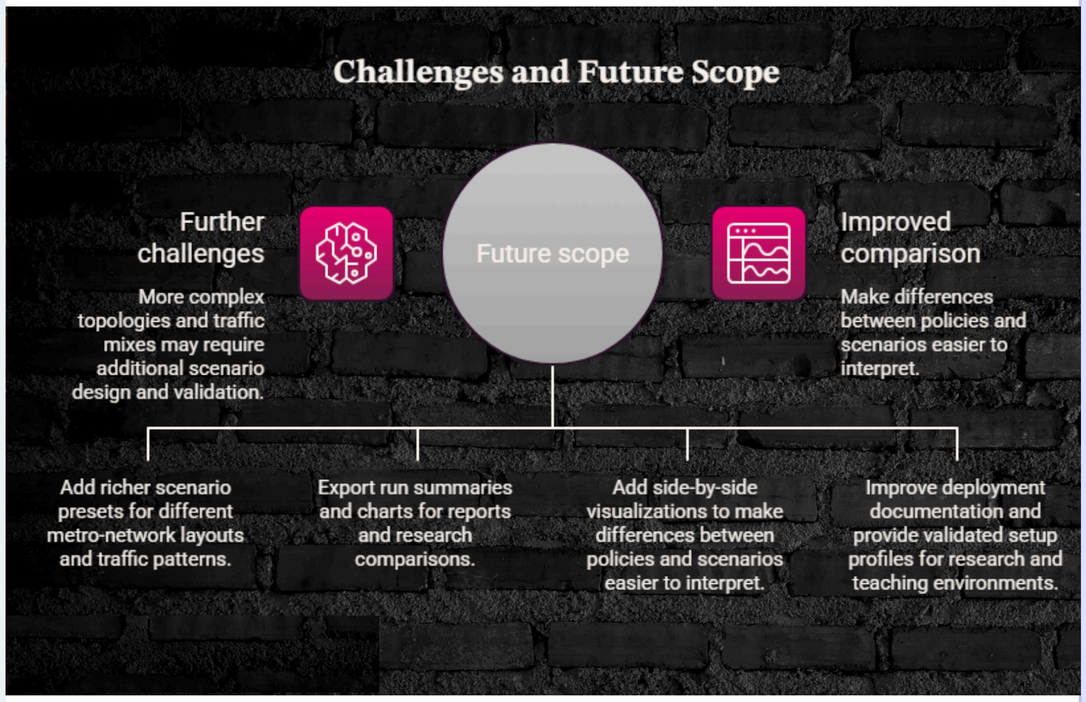
Challenges & Future Scope

Current Challenges

- Link physics are illustrative and have not been calibrated against a specific quantum-network experiment.
- PTP delay-model validation covers the available SLATE-PTP runs; performance on other setups is unknown.
- Simulations run as in-process background tasks, which limits long-running or large-scale workloads.
- The SeQUeNCe adapter is currently an unavailable stub.

Future Scope

- Calibrate link models with experimental data and validate them against physical testbeds.
- Expand PTP datasets and evaluate model performance on more network configurations.
- Add a durable job queue to support longer simulations and concurrent workloads.
- Implement and validate a SeQUeNCe integration for broader network studies.

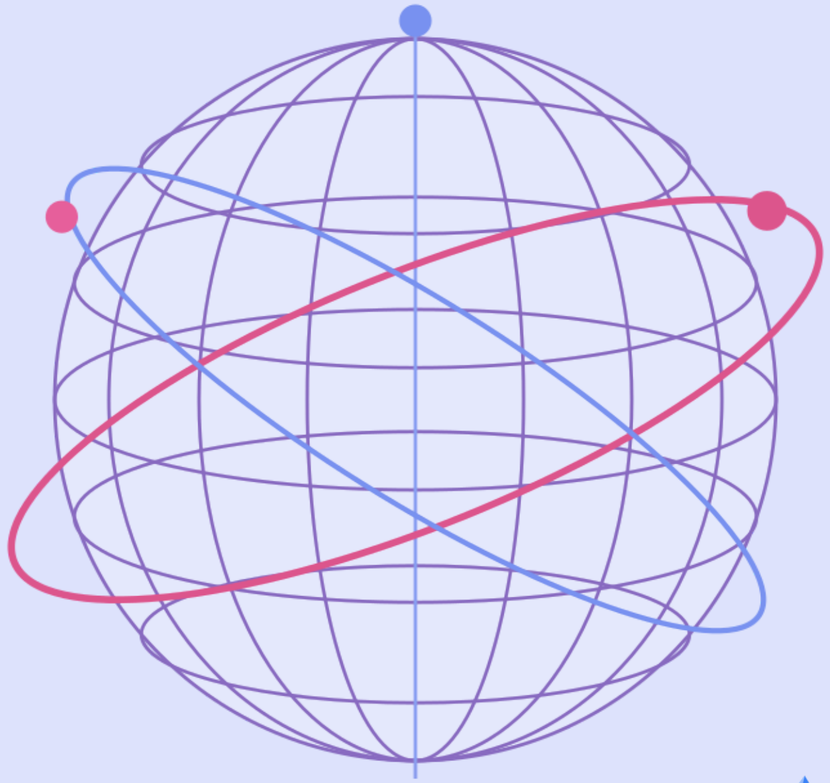


CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-quant-rangers

DEMO VIDEO LINK
https://youtu.be/gMAENA4j8C4